# Machine Learning Portfolio

Applied AI in Technology | Sampada Pote

## Portfolio Overview

This portfolio documents seven machine learning projects completed across the Applied AI in Technology course. Each project corresponds to a chapter from Géron's Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow (3rd ed., 2022) and was built using Google Collab. Together, they represent a complete journey through the machine learning lifecycle, from foundational concepts and classical algorithms to deep learning and natural language processing.

The projects progress deliberately in complexity: beginning with understanding what machine learning is and how pipelines work, moving through increasingly sophisticated algorithms, and culminating in neural networks and language models. Each notebook includes working code, documented reasoning, and reflection on what worked, what did not, and why. AI assistance was used in select steps and is logged within each notebook, with all outputs independently verified.

## Project Summary

| # | Topic | Techniques & Concepts Covered |
|---|---|---|
| Project 1 | ML Review (Ch. 1) | End-to-end classification pipeline; supervised, unsupervised & reinforcement learning; train/test split; cross-validation; baseline model comparison |
| Project 2 | Classification (Ch. 3) | Binary & multi-class classification; logistic regression; k-NN; confusion matrix; precision, recall, F1-score; ROC curves; decision thresholds |
| Project 3 | Support Vector Machines (Ch. 5) | Linear & non-linear SVMs; hyperplane optimization; kernel trick (RBF, polynomial); C and gamma tuning; high-dimensional classification |
| Project 4 | Decision Trees (Ch. 6) | Tree construction using Gini impurity & entropy; information gain; depth-limiting to control overfitting; pruning; visualization of decision paths |
| Project 5 | Dimensionality Reduction (Ch. 8) | Principal Component Analysis (PCA); variance retention; Kernel PCA for non-linear data; feature compression; visualization of reduced feature spaces |
| Project 6 | Neural Networks (Ch. 10) | Feedforward ANN architecture; forward propagation; backpropagation; activation functions (ReLU, sigmoid); overfitting diagnosis; learning curves in Keras/TensorFlow |
| Project 7 | Natural Language Processing (Ch. 16) | Text preprocessing pipeline; tokenization; stop word removal; stemming & lemmatization; TF-IDF; sentiment classification; transformer architecture overview |

## Skills Developed

Working through these projects built a coherent, layered skill set across four areas:

Data preparation and preprocessing. Before any model is trained, data must be cleaned, normalized, and shaped into a form the algorithm can learn from. Across every project, this step proved to be the most consequential. Handling missing values, encoding categorical variables, scaling features, and splitting data correctly into training and test sets were repeated consistently throughout the course, building strong practical habits around data integrity.

Model selection and evaluation. Each project required choosing an appropriate algorithm, understanding why it fits the problem, and measuring its performance honestly. Rather than relying on accuracy alone, the course-built fluency with a wider evaluation toolkit: precision, recall, F1-score, confusion matrices, ROC curves, and learning curves. These metrics matter because a model that looks accurate on average can still fail systematically for specific groups or cases.

Hyperparameter tuning and diagnostics. Understanding the difference between underfitting and overfitting and knowing which direction a model is failing in is one of the most practical skills developed in this course. Techniques such as cross-validation, depth limits in decision trees, regularization in SVMs, and dropout in neural networks all address the same underlying challenge: building models that generalize to new data rather than simply memorizing training examples.

Feature engineering and representation. The dimensionality reduction and NLP projects in particular highlighted how much the representation of data shapes what a model can learn. PCA showed how to compress high-dimensional feature spaces without losing essential information. The NLP pipeline (tokenization, stop word removal, stemming, TF-IDF) demonstrated that turning raw text into something a model can process is itself a discipline, not a preprocessing afterthought.

## Reflection on Learning

The most significant shift this course produced was in how the machine learning pipeline is understood. Before this course, it was easy to think of ML as a matter of picking a model and running it on data. What the course made concrete, through hands-on projects rather than lecture alone is that the work before the model (data preparation, feature selection, problem framing) and the work after the model (evaluation, diagnostics, honest reporting) matter at least as much as the algorithm itself.

The progression from logistic regression and k-NN through to neural networks and NLP also revealed something important: more complex models are not always better ones. A decision tree that can be visualized and explained often serves a real-world use case better than a deep network whose reasoning is opaque. Interpretability is not a limitation to work around, it is a design choice with real consequences for the people who rely on a model's outputs.

Designing and building in Google Collaboratory over the course of seven projects also developed a practical sense of reproducibility, how to structure a notebook so that someone else (or your future self) can run it, understand it, and trust its results. That discipline, more than any single algorithm, reflects the standard expected of professional ML work.

## Favorite Project: Natural Language Processing (Project 7)

Project 7 on Natural Language Processing stood out as the most conceptually surprising project in the portfolio. The idea that a machine can be trained to classify sentiment, identify entities, or generate coherent text by learning statistical patterns over massive amounts of human language, without ever being explicitly taught grammar or meaning challenged intuitions about what learning actually is.

The NLP pipeline required combining nearly every technique from earlier in the course: preprocessing (like the data pipeline work in Project 1), classification (from Project 2), and feature representation (from Project 5's dimensionality reduction). It felt like a synthesis of the whole course, and it made the jump from classical ML to transformer-based models which underlie tools like the ones used to assist in these very projects feel like a natural and logical progression rather than a leap into the unknown.

## How These Projects Prepare for Real-World ML Work

Real-world machine learning work is not primarily about knowing algorithms, it is about asking the right questions before building anything. What is the actual prediction task? Is the data clean and representative? What does a mistake cost? Who is affected by the model's outputs? These questions were woven into every project in this course, and that framing is what distinguishes thoughtful ML practice from simply running code.

The seven projects together demonstrate an ability to take a problem from raw data to evaluated model, document decisions along the way, and reflect honestly on limitations and tradeoffs. That end-to-end fluency, combined with a growing awareness of the ethical dimensions of model deployment is what the course set out to build, and what this portfolio demonstrates.

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## Reference

Géron, A. (2022). Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow (3rd ed.). O'Reilly Media.